

Save and Load — JSON Game State

Day 03

Lesson 03

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Course 02

Phase 1 — Game Studio

Time: 60–75 min

Mode: Self-study

Today's goals:

- ★ Save player HP and inventory to a JSON file
- ★ Load a saved game on startup
- ★ Offer continue or new game



Course 02 quest map: Day 03 of 18

Hook

Real games let you quit and come back. Today your dragon fight gets a save slot.

Learn



Learn

`json.dump` writes; `json.load` reads. Save `player_hp`, `dragon_hp`, `poison_uses`. Always use `with open(...)`. If no save file, start a new game.



AI rules today

Predict/Run: AI off. **Investigate:** Socratic hints only. **Make:** you review every line.

Predict — Run — Investigate — Modify —> Make



Demo code

Folder: day03/

```
1 import json
2 from pathlib import Path
3
4 SAVE = Path("save_slot.json")
5
6 def save_game(player_hp, dragon_hp):
7     with open(SAVE, "w") as f:
8         json.dump({"player_hp": player_hp, "dragon_hp": dragon_hp}, f, indent=2)
9
10 def load_game():
11     if not SAVE.exists():
12         return None
```

Full file: demos/demo_save_load.py

 Try it

In terminal, `cd course02/day03`, then predict output, then run: `python demos/demo_save_load.py`

 Assignment

Work in: `day03/assignment/`

- A — **Predict** If `save.json` does not exist yet, should load crash or start a new game?
- B — **Debug** `assignment/buggy_save.py` — file not closed / wrong mode.
- C — **Level up** Add save command and auto-load on start (with “new game” option).
- D — **Explain** Why is JSON better than copying numbers into comments?

 Checklist

- ☐ Re-read **Learn**; predict outputs (10 min)
- ☐ Assignments A–D (35–45 min)
- ☐ Run your program 3 times with different inputs
- ☐ Stuck 15 min → C-R-A-V (`../shared/prompts.md`)

Reflection

1. Easiest part today?
2. One line you can teach a friend.

Tomorrow: Log every fight to a CSV file.